Deformation analysis of the Alqueva dam using EGMS InSAR data

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Abstract

Understanding and predicting the behaviour of geotechnical structures is still a complex challenge. Traditional methods often fail to fully capture the complexities of these structures, which include challenges such as slope instabilities, dam behaviour, and landslides. To address these issues, significant advances have been made in Earth observation techniques, such as radar interferometry and satellite imagery, as well as in machine learning tools. These developments have provided new opportunities for monitoring and predictive modelling. This review explores the application of Earth observation techniques, such as InSAR, for detecting ground displacement and deformation. It also examines the role of machine learning in analysing large datasets, detecting patterns and anomalies, and predicting potential geotechnical failures. Despite these advancements, challenges remain, particularly regarding the reliance on high-quality data and the integration of advanced algorithms into practical applications.

1 Introduction

1.1 Contexto

Geotechnical structures, particularly large-scale concrete dams, play a critical role in global infrastructure, environmental management, and energy generation. Ensuring their structural integrity is paramount to preventing catastrophic failures. Traditionally, monitoring systems have focused on "observing the main actions and characterising the quasi-static structural response" (Mata et al., 2023). These methods, based on physical inspections and *in-situ* instrumentation, have long been the standard for ensuring stability. Since 1967, the Hydrostatic-Seasonal-Time (HST) model, first proposed by Willm and Beaujoint, has remained the most popular data-based behavior model in dam engineering (Salazar et al., 2017). However, as the scale and complexity of modern projects increase, these traditional methods have become "increasingly insufficient" (Hariri-Ardebili et al., 2023). A significant challenge lies in the reliability of instruments placed in "hostile environments, where humidity, temperature change, and freeze" put any mechanical or electronic tool to the test (De Sortis & Paoliani, 2007), making the case for more robust, remote-sensing-based frameworks, especially in remote areas.

(Mata et al., 2023) "Traditionally, monitoring systems of large concrete dams have been based on observing the main actions and characterising the quasi-static structural response.

(Hariri-Ardebili et al., 2023) "Traditional dam safety methods, based on visual inspections and manual monitoring, have long been the standard for ensuring the stability and safety of dams. However, as the scale and complexity of dam projects have increased, these methods have become increasingly insufficient."

(Salazar González et al., 2017) "Linear regression is the simplest statistical technique, appropriate to reproduce certain phenomena. It is also the basis of the most popular data-based behaviour model in dam engineering: the Hydrostatic-Season-Time (HST). It was first proposed by Willm and Beaujoint in 1967."

(Ruiz-Armenteros et al., 2023) "Periodic in-person inspections by specialized technicians are conducted to maintain the facilities, ranging from daily to annual visits.

(De Sortis & Paoliani, 2007) "Actually it is not easy to get reliable measures by means of instruments often placed in hostile environments, where humidity, temperature change, freeze and uncomfortable access put to the test any kind of mechanical and electronic tool." isto pode ser útil para justificar a framework e ser utilizada em áreas mais remotas

1.2 Avanços Insar e EO

Recent advancements in Multi-Temporal InSAR (MT-InSAR) have revolutionized geotechnical monitoring, offering millimetre-level precision regardless of lighting or weather conditions (Cernuto et al., 2026; Xie et al., 2022). Within the InSAR family, specialized approaches such as Persistent Scatterer (PSInSAR) and Small Baseline Subset (SBAS) have gained prominence for their ability to detect subtle ground movements (Hariri-Ardebili et al., 2020). However, structural monitoring of dams using MT-InSAR is not yet a widespread application (Ruiz-Armenteros et al., 2018). Nevertheless, previous achievements suggest these techniques can support warning systems by adding redundancy at a low cost.

1.3 Oportunidade

A major technological opportunity emerged in 2022 with the launch of the European Ground Motion Service (EGMS). Implemented by the EEA and ESA, it provides standardized, open-access displacement data that complements on-site monitoring by offering "broader spatial coverage and continuous monitoring capabilities" (Ruiz-Armenteros et al., 2023). By examining the temporal patterns in EGMS time series, it is possible to identify long-term trends or seasonal variations critical for assessing risks. Simultaneously, Artificial Intelligence (AI) models are becoming increasingly sophisticated, opening "fresh opportunities for dam's digital and intelligent development" (Cao et al., 2025).

(Ruiz-Armenteros et al., 2023) "In addition to on-site monitoring techniques, the EGMS complements these methods by offering broader spatial coverage and continuous monitoring capabilities. It assists in identifying areas of interest for further investigation and validation, allowing dam operators and authorities to allocate their resources effectively."

(Ruiz-Armenteros et al., 2023) "The EGMS generates time series data that illustrate the evolution of ground motion over a specific period. By examining the temporal patterns of ground displacement, it becomes possible to identify longterm trends or seasonal variations in the dam's behavior and its surroundings. This information is valuable for understanding the dam's stability and assessing any changes that may pose a risk."

(Cao et al., 2025) "After decades of research, AI in particular are becoming increasingly sophisticated. Theses open up fresh opportunities for dam's digital and intelligent development (Sun et al., 2018; Zhang et al., 2019)."

1.4 Estudos prévios em barragens

Several studies have explored InSAR displacement series for dam monitoring. Projects like SIAGUA in Spain have focused on using Copernicus data as routine tools for dam surveillance (Ruiz-Armenteros et al., 2023). Methodologically, new tools such as SARClust have introduced semi-automatic clustering of InSAR time series to group points with "homogeneous behavior" (Roque et al., 2023). However, many of these efforts focus on retrospective analysis rather than continuous operational safety frameworks.

(Roque et al., 2023) "In this study, a new tool for semiautomatic analysis of InSAR displacement time series is presented, called SARClust—Clustering of InSAR displacement time series. The tool forms clusters of InSAR measurement points with homogeneous behavior. This means the points in each cluster present similar displacement time series among themselves, but different from the displacement time series of the points in other clusters."

(Ruiz-Armenteros et al., 2023) "The SIAGUA project, funded by the Spanish Ministry of Science and Innovation, focuses on applying the Copernicus land monitoring program and other satellite radar sensors as routine tools for monitoring dams, large ponds, and their surroundings. EGMS is used to provide initial deformation data to dam managers, enabling surveillance of the infrastructures under their responsibility."

(Ruiz-Armenteros et al., 2023) "The SIAGUA project in Spain utilizes EGMS to monitor dams and surrounding areas, providing surveillance information to dam managers."

(Ruiz-Armenteros et al., 2018) "Structural monitoring of dams using MT-InSAR is not yet a widespread application and only a relative few number of studies have been conducted during the past 15 years for structural health monitoring of dams using SAR data12-24. However, these achievements allowed to conclude that MT-InSAR techniques have the potential to support the development of new and more effective means of monitoring and analyzing the health of dams and add redundancy, at low cost, to support and assist warning systems."

1.5 Limitações (referidas por outros autores)

Despite these benefits, practical integration remains challenging. Machine Learning (ML) models, though often more capable than HST of reproducing non-linear effects, typically focus on specific case studies or arch dam typologies, which represent only 5% of dams worldwide (Salazar et al., 2021). Furthermore, the technical nature of radar data presents a barrier; the "practical utility of solely LOS (Line-of-Sight) deformation is limited for users unfamiliar with InSAR," often leading to confusion among dam managers accustomed to horizontal and vertical displacement vectors (Ruiz-Armenteros et al., 2023). Additionally, users must be cautious of EGMS data accuracy, as discrepancies compared to ground truth can sometimes reach 5–10 mm/yr due to atmospheric or geometric artifacts (Crosetto et al., 2025).

(Salazar et al., 2021) "There is a growing interest in the application of innovative tools in dam monitoring data analysis. Although only HST is fully implemented in engineering practice, the number of publications on the application of other methods has increased considerably in recent years, specially NN. It seems clear that the models based on ML algorithms can offer more accurate estimates of the dam behaviour than the HST method in many cases. In general, they are more suitable to reproduce non-linear effects and complex interactions between input variables and dam response. However, most of the published works refer to specific case studies, certain dam typologies or determined outputs. Many focus on radial displacements in arch dams, although this typology represents roughly 5 of dams in operation worldwide."

(Salazar et al., 2021) ”From a practical viewpoint, data-based models should also be user-friendly and easily understood by civil engineering practitioners, typically unfamiliar with computer science, who have the responsibility for decision making.”

(Salazar et al., 2017) ”The most popular data-based approach for dam monitoring analysis is the hydrostatic-seasonal-time (HST) model. It was first proposed by Willm and Beaujoint in 1967 [9] to predict displacements in concrete dams, and has been widely used since.”

While EGMS provides a continental-scale baseline, its application to specific infrastructure like dams must account for inherent InSAR limitations, such as...:

(Crosetto et al., 2025), ”Some of the results show a good agreement between EGMS and the ground truth, both in terms of deformation velocity (absolute differences below 1–2 mm/yr) and time series (standard deviation of differences below 5 mm). However, this cannot be said generally. In some cases, the differences in the deformation velocities (EGMS vs. ground truth) are far from the millimetre-per-year precision, which is sought. For instance, differences in the range of 5–10 mm/yr can be found. The same is true for the time series, where biases up to 10–20 mm or peaks of the same order of magnitude occur.”

(Ruiz-Armenteros et al., 2023) ”However, utilizing InSAR technology requires specialized knowledge of various processing algorithms and significant computational resources.”

1.6 Research Gap

A critical research gap persists where satellite remote sensing, ML, and traditional monitoring operate as ”isolated islands.” A technical bottleneck in the current EGMS Ortho products—which represent vertical and east-west deformation—is their 100-meter resample, which is ”too large to accurately characterize local movements” in specific dams (Ruiz-Armenteros et al., 2023). This necessitates a framework that can combine ascending and descending geometries to overcome the LOS limitation. Moreover, there is an urgent need for models that are ”user-friendly and easily understood by civil engineering practitioners,” bridging the communication gap between computer science and dam safety (Salazar et al., 2021).

(Ruiz-Armenteros et al., 2023) ”However, the Ortho products, which represent the vertical and east-west deformation, are not useful for this dam due to its relatively small size. A 100meter resample is too large to accurately characterize the vertical and east-west deformation patterns, as it does not capture local movements. Moreover, the Ortho products from the EGMS are not suitable for comparison with on-site measurements such as leveling over the dam’s crest or berms.” isto é muito importante para justificar a importância da combinação das órbitas ascendente e descendente

(Ruiz-Armenteros et al., 2023) ”While the EGMS is highly beneficial for dam managers, the practical utility of solely LOS deformation is limited for users unfamiliar with InSAR, as the deformation is projected in the LOS direction. Dam managers are accustomed to working with deformations in the horizontal and vertical directions, and the deformation in the LOS direction can lead to confusion.

1.7 Solução

To address this challenge, this paper proposes the SAFE-DAM (Satellite Automated Framework for Engineering of Dams). This framework integrates EGMS data with Machine Learning clustering algorithms to transform massive satellite datasets into operational safety indicators. Unlike purely theoretical studies, this work validates the SAFE-DAM methodology using a decades-long dataset from the Alqueva

Dam. By cross-referencing InSAR-derived clusters with *in-situ* pendulum measurements, this research demonstrates how satellite monitoring can complement traditional methods, ensuring that ”engineering judgement based on experience” remains central to interpreting automated results (Salazar et al., 2021).

2 Methodology

(Cernuto et al., 2026) ”To quantify the interferometric coverage of landslide phenomena, each landslide polygon was divided into a regular grid of square cells, each measuring 25 metres per side. This grid size was chosen as a compromise between two contrasting needs: on one hand, a grid that is too large may lead to an overly generalised representation, where a few interferometric points cover extensive portions of the landslide, resulting in a loss of spatial detail; on the other hand, an excessively fine grid increases the likelihood of cells containing no data, due to the limited density of PSs in the EGMS datasets. The analysis was carried out separately for each orbital geometry (ascending and descending).”

(Cernuto et al., 2026) ”each landslide polygon was discretised using square cells of 25×25 m. Each cell was assigned a representative 206 value corresponding to the median of the LOS velocities of the PSs contained within it (Figs. 5a, b). The use of the median 207 provides a robust estimate of the local central tendency, reducing the influence of outliers or typical InSAR-related disturbances, 208 such as atmospheric noise, anomalous reflections, or phase discontinuities. Once median velocity values were assigned to 209 each cell, it became possible to describe the internal deformation pattern of each landslide polygon through a histogram of the 210 cell-based median velocities, providing an empirical representation of the spatial variability of the phenomenon.”

(Cernuto et al., 2026) ”Processing ascending and descending geometries separately proved essential 417 to highlight the influence of viewing direction and the complementarity between datasets: phenomena poorly visible in one 418 geometry were often detectable in the other, confirming that integrating both perspectives reduces shadow zones and provides a 419 more complete picture of interferometric observability.”

(Foroughnia et al., 2025) ”A well-established method that provides building-level information is Multi-Temporal Interferometric SAR (MT-InSAR), which uses stacks of SAR images. Thanks to the all-weather image acquisition capability of SAR sensors and their ability to cover large areas with short revisit times (Macchiarulo et al., 2024), MT-InSAR can monitor extensive regions affected by earthquakes using frequent SAR data. Additionally, high-resolution SAR data can be freely available through open data programs (ESA, 2024), making it well suited to accurate building-level assessments (Macchiarulo et al., 2024). The MT-InSAR technique identifies stable points, known as Persistent Scatterers (PSs), by analysing stacks of SAR images taken over time from the same area. These PSs typically correspond to man-made objects such as buildings, enabling detailed monitoring of changes at the building level. MTInSAR has been widely used for long-term displacement monitoring in urban areas (Ciampalini et al., 2014; Bianchini et al., 2015; Foroughnia et al., 2019). However, no study has yet explored this technique to assess post-earthquake building reconstruction, nor has it been utilised to support field investigations.”

2.1 Displacements calculation and validation

(Ruiz-Armenteros et al., 2023) ”However, the Ortho products, which represent the vertical and east-west deformation, are not useful for this dam due to its relatively small size. A 100meter resample is too large to accurately characterize the vertical and east-west deformation patterns, as it does not capture local movements. Moreover, the Ortho products from the EGMS are not suitable for comparison with on-site measurements such as leveling over the dam's crest or berms." isto é muito importante para justificar a importância da combinação das órbitas ascendente e descendente

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The EGMS data used in this study include both ascending and descending Line-of-Sight (LOS) products, which represent ground displacements along the radar viewing direction. These LOS measurements were decomposed into vertical (Up–Down) and horizontal (East–West) components, enabling direct comparison with reference displacements derived from orthophotogrammetry (hereafter referred to as "Ortho").

Additionally, the study employed a Digital Elevation Model (DEM) and related metadata to account for topographic effects and to support geometric corrections. The combination of Calibrated (ascending and descending) EGMS products with ORTHO displacements allows for a comprehensive evaluation of both the vertical and horizontal deformation behavior of the structure. This integrated approach provides valuable insights into the dam's mechanical response and contributes to the validation of satellite-based ground motion measurements in complex geotechnical settings.

Figure 1: Views of the Alqueva Dam from different perspectives.

2.1.1 Decomposition of Line-of-Sight displacements

To derive the vertical (dV) and horizontal (dH) displacement components over the Alqueva dam, a dedicated processing workflow was implemented using Python and geospatial libraries (Pandas, GeoPandas, NumPy, SciPy). The procedure combines calibrated ascending (ASC) and descending (DESC) InSAR datasets from the EGMS Level-2b product and validates the decomposed components against the orthogonal vertical (U) and east–west (E) displacements provided by the EGMS Level-3 (ORTHO) product.

Data reading and spatial filtering
The calibrated ASC, DESC, and ORTHO datasets were imported into Python and spatially filtered to the dam and its immediate surroundings, defined by fixed coordinate limits in the ETRS89-LAEA (EPSG:3035) projection. This ensured consistent spatial referencing across all datasets and reduced the computational load to the relevant monitoring area.

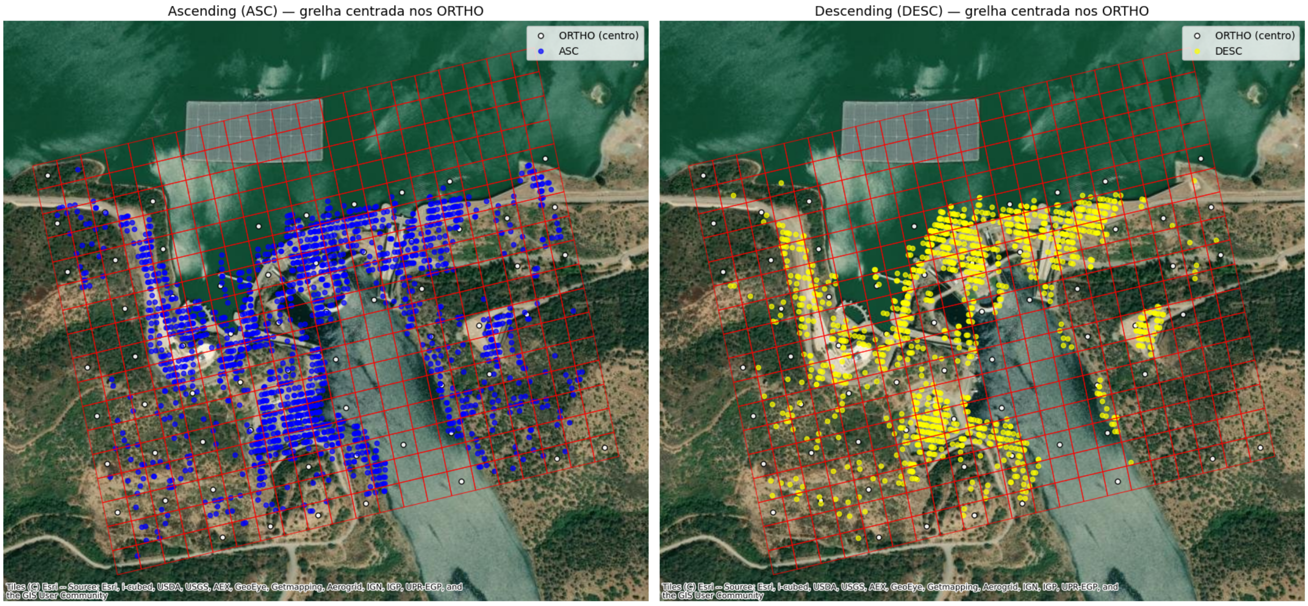


Figure 2: View of the dam and initial ASC/DESC point distribution maps illustrate the area of interest and the spatial coverage of the datasets.

Time-series reshaping

The displacement columns in the ASC, DESC, and ORTHO datasets were reshaped into a long-format table, with one row per spatial point and acquisition date. This format facilitated temporal interpolation and subsequent synchronization of datasets acquired on different dates.

Temporal interpolation

Since ASC and DESC acquisitions are collected on distinct dates, a common set of monthly timestamps was defined. Linear interpolation was applied to generate displacement values for both orbits at the same epoch. This ensured temporal consistency and enabled direct comparison between ASC and DESC measurements.

Spatial interpolation (IDW)

Since ASC and DESC pixels are not spatially coincident, the DESC displacements were interpolated onto the ASC grid using Inverse Distance Weighting (IDW). For each ASC point, the interpolation used the five nearest DESC neighbours within a 150 m radius, weighted by the inverse square of their distance. This step also interpolated the incidence angle ( ) and track angle ( ) from the DESC geometry, ensuring that both ASC and DESC values correspond to the same ground location.

Computation of geometric parameters (β and γ)

Two auxiliary geometric parameters are defined for each interpolated ascending (ASC) point:

- β (orbital-geometry term) — computed as:

$$\beta = \arcsin\left(\cos(I) \cdot \cos(\text{latitude})\right) \tag{1}$$

  where $I = 98.6°$ is the Sentinel-1 orbit inclination. In this work, β is used as a geometric correction term derived from the satellite orbital inclination and the point latitude. It is not a direct measurement of terrain slope aspect (i.e., not DEM-derived). It accounts for the relative orientation between the satellite orbit plane and the local observation latitude, consistent with the trigonometric formulation implemented in the processing code.

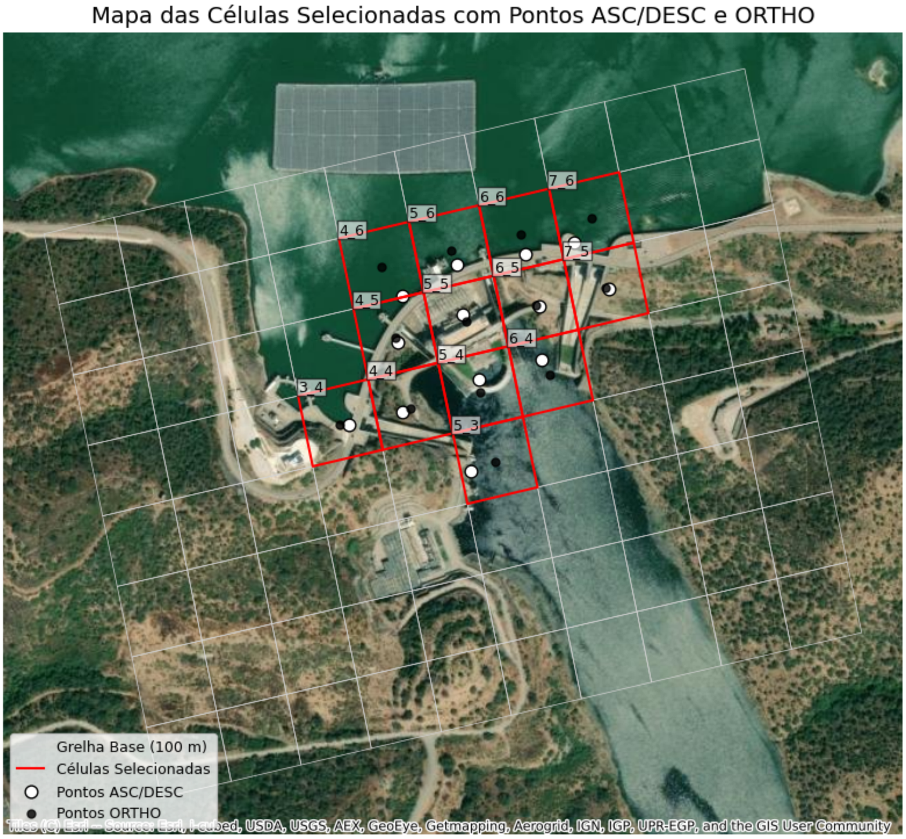


Figure 3: VMap of grid cells and points after IDW interpolation, showing the spatial alignment of ASC and DESC points.

- γ (cross-track tilt) — represents the local topographic tilt perpendicular to the radar line-of-sight. Given the relatively flat topography of the dam area, γ was set to zero ($\gamma = 0$), which simplifies the decomposition while introducing negligible error for this site.

Note: If a DEM-based local slope/aspect correction were applied, β would be replaced or augmented by the DEM-derived slope aspect, and γ by the DEM-derived cross-track tilt. In this study, those DEM corrections were not used.

Decomposition of Line-of-Sight (LOS) displacements into vertical and horizontal components

(Ruiz-Armenteros et al., 2023) "In this project, we estimate the vertical and east-west displacements by using Detektia's decomposition software, which is based on the nearest neighbors decomposition. Taking the positions of one of the geometries as a reference, each point is paired with the nearest point in the other geometry. The only compliance that they have to fulfill is that the paired points are not separated more than a certain predefined distance threshold."

Let d_{ASC} and d_{DESC} be the LOS displacements from ascending and descending passes, respectively. Let θ_A, θ_D be the local incidence angles (angle between LOS and vertical) for ASC and DESC, and α_A, α_D their respective orbit azimuth (track) angles.

The LOS measurements relate to the vertical and east–west horizontal components via:

$$\begin{cases} d_{\text{ASC}} = -d_V \cos\theta_A + d_H \sin\theta_A \cos(\alpha_A + \beta + \gamma) \\ d_{\text{DESC}} = -d_V \cos\theta_D + d_H \sin\theta_D \cos(\alpha_D + \beta + \gamma) \end{cases} \tag{2}$$

Solving this 2×2 trigonometric system yields the vertical and horizontal components:

$$d_V = \frac{d_{\mathrm{DESC}} \sin\theta_A \cos(\beta-\gamma) + d_{\mathrm{ASC}} \sin\theta_D \cos(\beta+\gamma)}{\cos\theta_A \sin\theta_D \cos(\beta-\gamma) + \cos\theta_D \sin\theta_A \cos(\beta+\gamma)} \tag{3}$$

$$d_H = \frac{d_{\mathrm{DESC}} \cos\theta_A - d_{\mathrm{ASC}} \cos\theta_D}{\cos\theta_A \sin\theta_D \cos(\beta-\gamma) + \cos\theta_D \sin\theta_A \cos(\beta+\gamma)} \tag{4}$$

These expressions match the implementation used in the processing code (with $\gamma = 0$ for this study).

Assumptions and implications:

- β interpretation: β is defined as an orbital-geometry term computed from orbit inclination and latitude (not DEM slope aspect). This matches the code implementation and avoids confusion with terrain aspect.
- $\gamma = 0$: Assuming zero cross-track tilt is reasonable for the relatively flat dam area. For steep abutments, γ should be estimated from a DEM.
- IDW interpolation (DESC $\rightarrow$ ASC): Spatial interpolation introduces estimation error where data are sparse. The chosen radius and neighbour selection mitigate this, but coherence masks and spatial distributions should be inspected.
- Sensitivity: Radar LOS is more sensitive to vertical motion than to east–west horizontal motion. Thus, d_V estimates are more robust, while d_H typically exhibits larger uncertainties.

Table 1: Summary of symbols and parameters used in the LOS decomposition.

Symbol	**Meaning**	**Source in code**	**Comment**
d_{ASC}, d_{DESC}	Line-of-sight (LOS) displacements from ascending and descending orbits	`disp`, `disp_idw` columns	Original InSAR data (ASC/DESC)
θ_A, θ_D	Incidence angle (between LOS and vertical)	`incidence_angle`, `theta_desc`	Retrieved from EGMS metadata
α_A, α_D	Azimuth angle of the satellite orbit (track angle)	`track_angle`, `alpha_desc`	Also obtained from EGMS metadata
β	Orbital geometric correction term (simplified)	`np.arcsin(...)` function	Not the surface aspect, but an approximate geometric factor
γ	Cross-track surface tilt	Set to 0	Assumes flat terrain
d_V, d_H	Vertical (U) and horizontal (E–W) displacements	Computed by trigonometric decomposition	Final results, compared with ORTHO data

This formulation ensures that displacements are correctly projected onto the vertical and horizontal reference frame, considering local terrain geometry. The decomposition was applied to each point within the dam area and aggregated into 100 m grid cells for subsequent comparison with the orthogonally derived displacements from the EGMS ORTHO products.

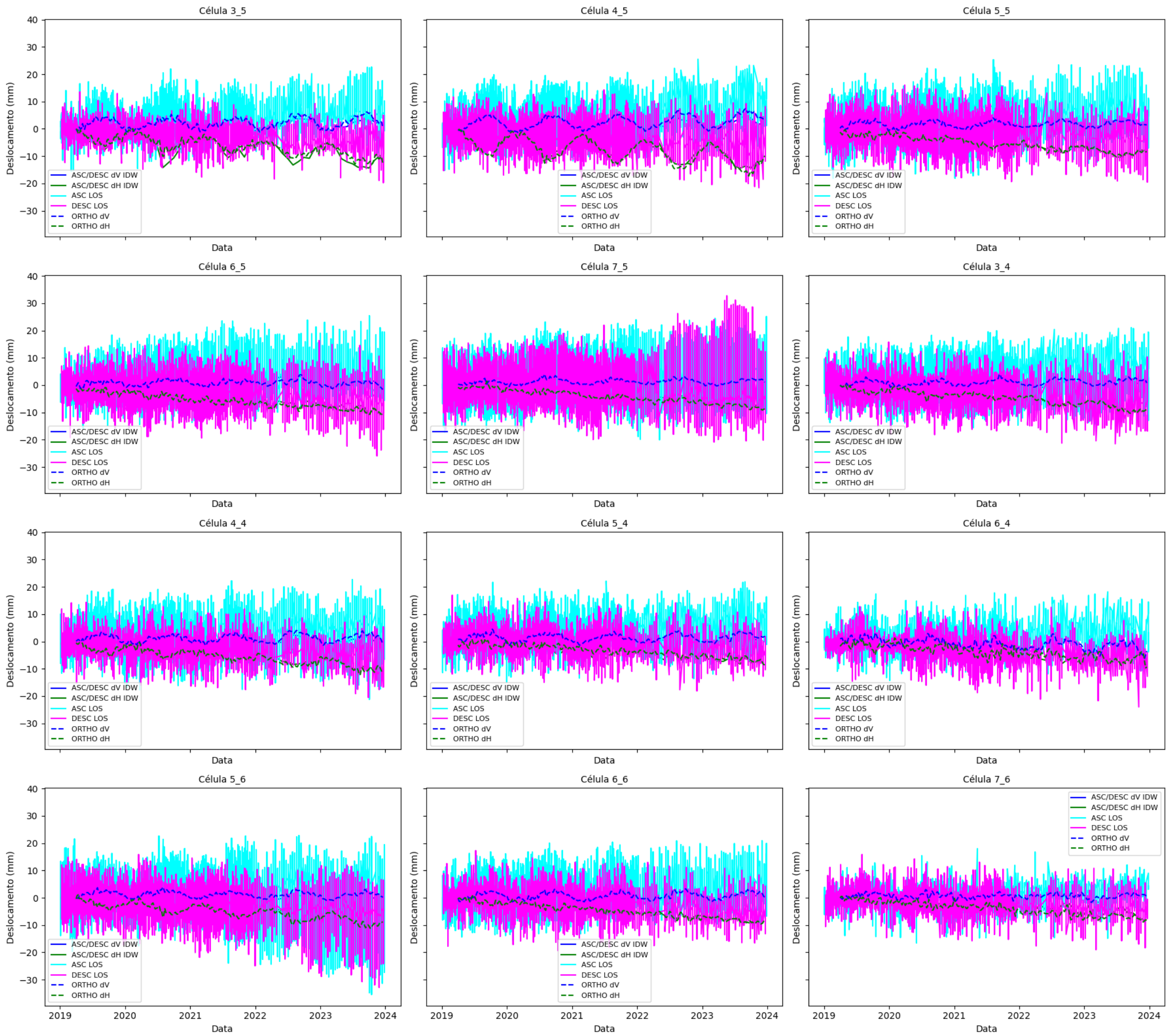


Figure 4: Extended plots including original ASC and DESC LOS measurements.

Spatial aggregation into grid cells

The computed dV and dH components were aggregated into 100 m grid cells centered on the ORTHO points. This spatial aggregation ensured consistency between InSAR-derived components and independent orthogonal measurements and facilitated comparison across the dam body.

Visualization and validation

Selected cells across the dam were analyzed in detail. For each cell, vertical and horizontal displacement time-series were plotted, comparing ASC/DESC LOS measurements, ASC/DESC IDW decomposed components, and ORTHO displacements.

2.1.2 Validation with orthophoto-derived displacements

To evaluate the reliability of InSAR-derived displacements, both d_V and d_H components were compared to independent measurements obtained from orthophoto data.

For each displacement direction, pairs of co-located observations were extracted, resulting in two datasets:

- d_V^{InSAR} vs. d_V^{Ortho}

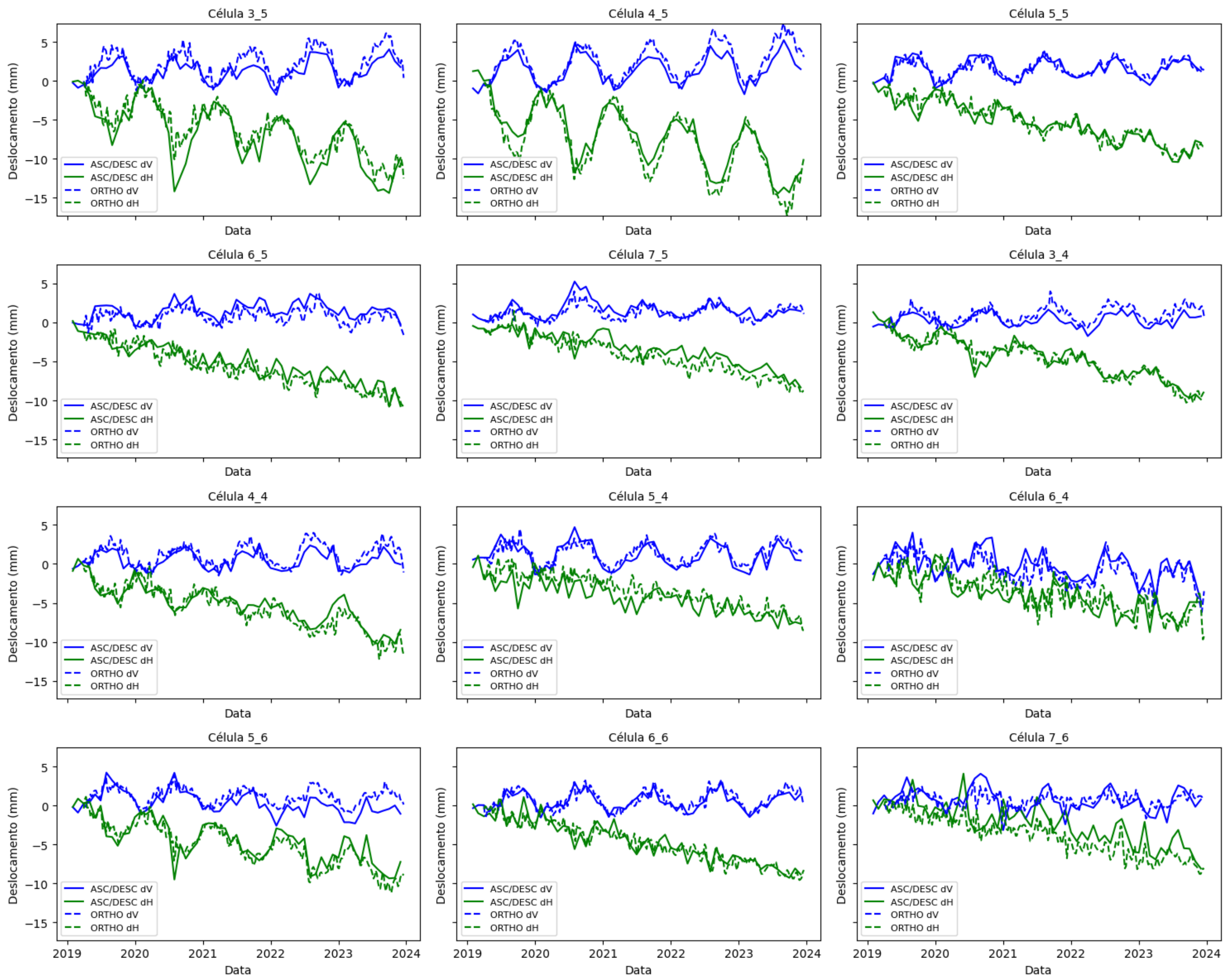


Figure 5: Time-series plots of dV and dH from ASC/DESC IDW compared to ORTHO.

- d_H^{InSAR} vs. d_H^{Ortho}

These datasets were analysed through scatter plots, and their correspondence was quantified using standard statistical metrics.

2.1.3 Accuracy assessment metrics

The consistency between InSAR and Ortho displacements was evaluated using RMSE, MAE, and Pearson's r, measuring respectively the average error, absolute deviation, and linear correlation between datasets.

- Root Mean Square Error (RMSE):

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{5}$$

  which measures the average magnitude of the error in millimetres.

- Mean Absolute Error (MAE):

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{6}$$

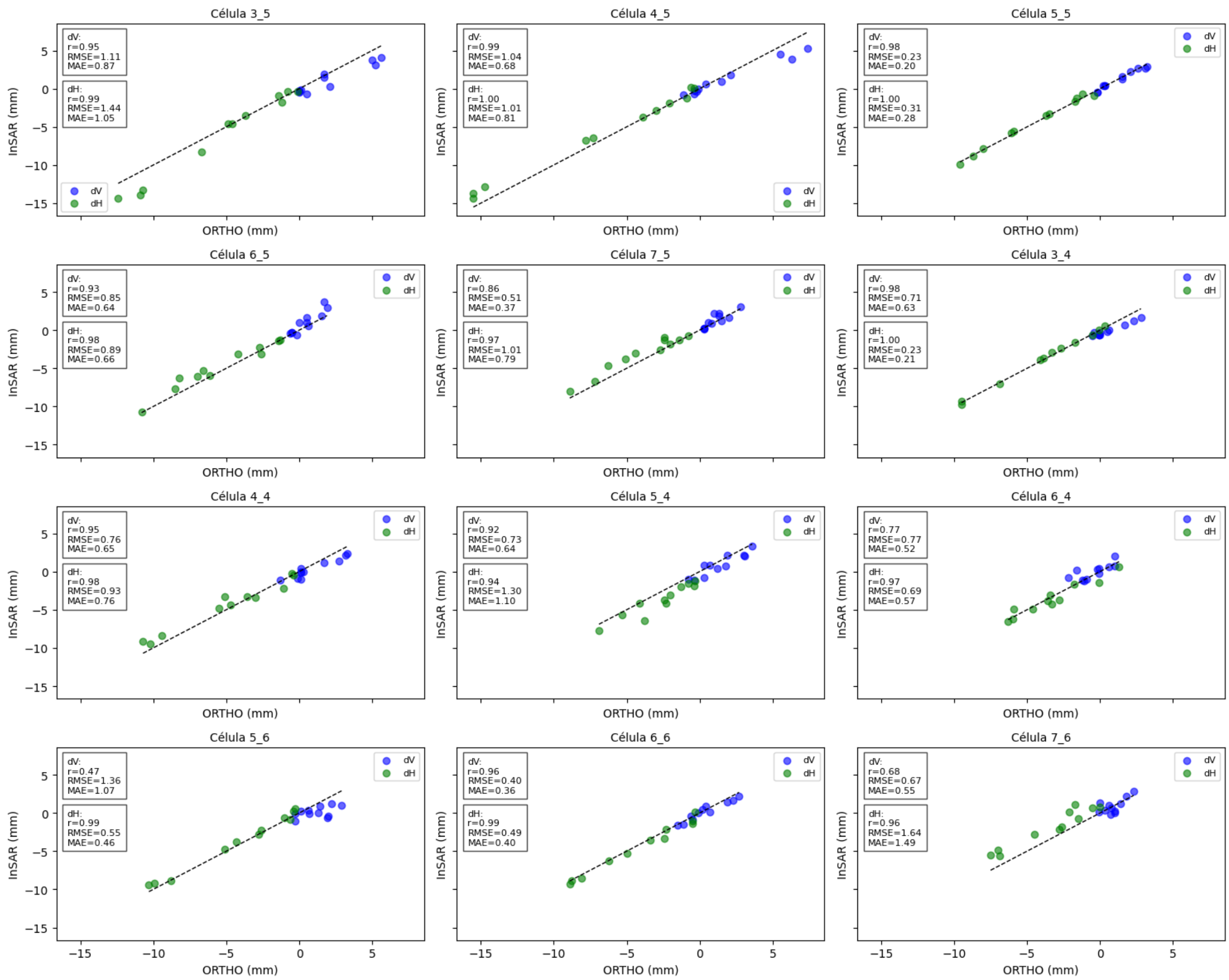


Figure 6: Plots showing line-identity comparison and statistical validation metrics.

- Pearson Correlation Coefficient (r):

$$r = \frac{\mathrm{cov}(y_i, \hat{y}_i)}{\sigma_y \, \sigma_{\hat{y}}} \tag{7}$$

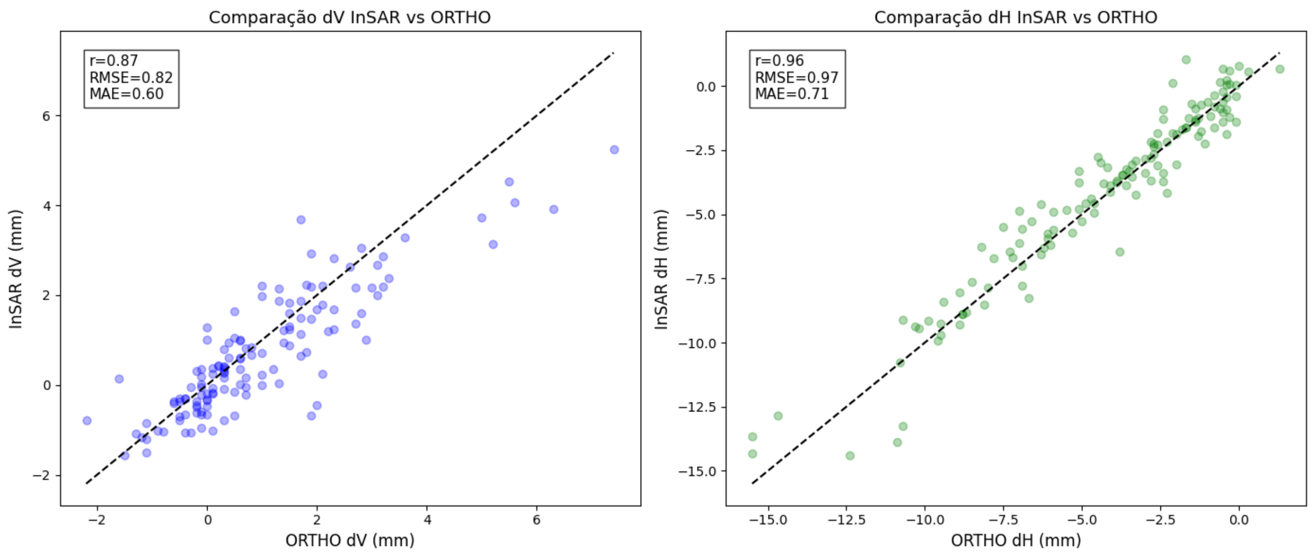


Figure 7: ....

2.2 Clustering and Pattern Analysis

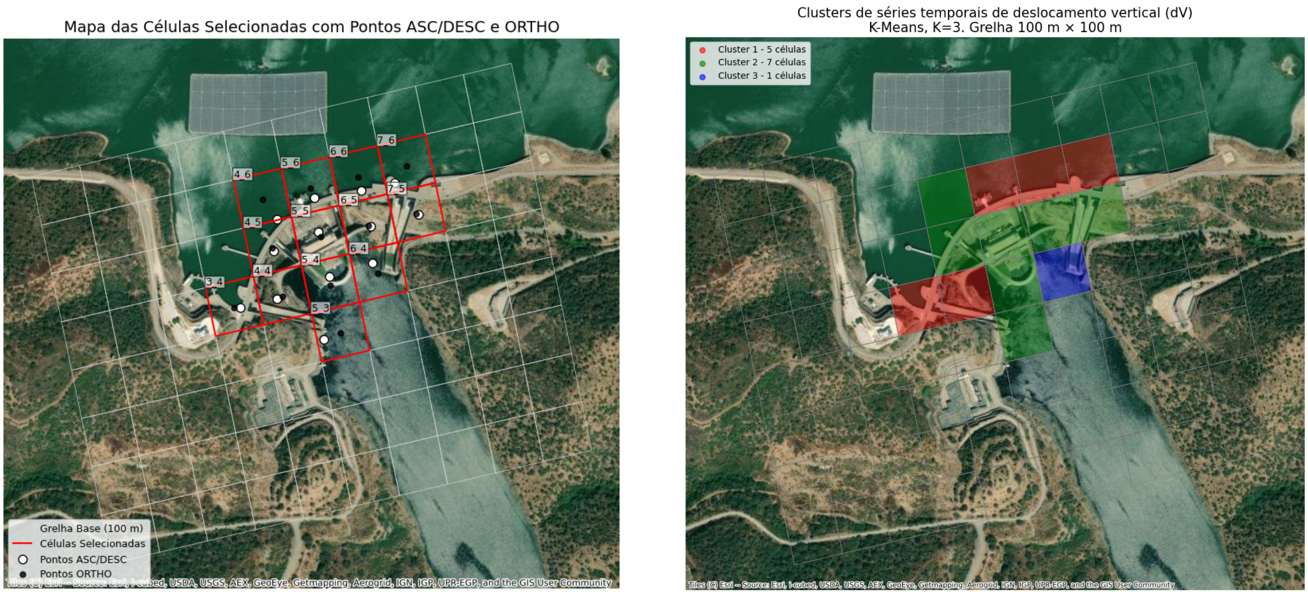
2.2.1 overview

Clustering techniques were applied to identify spatial and temporal patterns of deformation within the InSAR-derived displacement datasets. These unsupervised machine learning methods enable the grouping of areas with similar deformation behaviour, facilitating the identification of zones with homogeneous motion, potential structural anomalies, or early indicators of instability.

In this study, both spatial clustering (using average displacement values) and temporal clustering (using time-series of deformation) were implemented. The main algorithms adopted were K-Means and Hierarchical Clustering using DTW distance.

2.2.2 K-Means Clustering

The K-Means algorithm was first employed to partition the InSAR displacement time-series into a predefined number of clusters based on their overall temporal similarity. This method minimizes intra-cluster variance by iteratively assigning each time-series to the nearest cluster centroid and recalculating centroids until convergence. The optimal number of clusters kwas determined through the Silhouette Score and the Elbow Method, ensuring a balance between cluster compactness and separation. K-Means provided an efficient way to identify dominant deformation behaviours—such as stable, subsiding, or uplifting areas—serving as a first-level classification of the dataset.



(a) Study Area: Grid cells and InSAR observations. White points show the ortho points; colored points represent the processed ascending and descending orbit acquisitions combined.

(b) K-Means clustering results for vertical displacement (dV) time-series. Each color represents a cluster of grid cells with similar temporal deformation patterns.

Figure 8: Overview of the study area and cluster analysis.

2.2.3 Hierarchical Clustering using DTW Distance

To complement the partition-based results of K-Means, Hierarchical Clustering using Dynamic Time Warping (DTW) distance was applied to further explore the nested relationships among deformation time-series. Unlike conventional Euclidean metrics, DTW accounts for time shifts and non-linear temporal variations, making it particularly suited for deformation signals influenced by seasonal or delayed

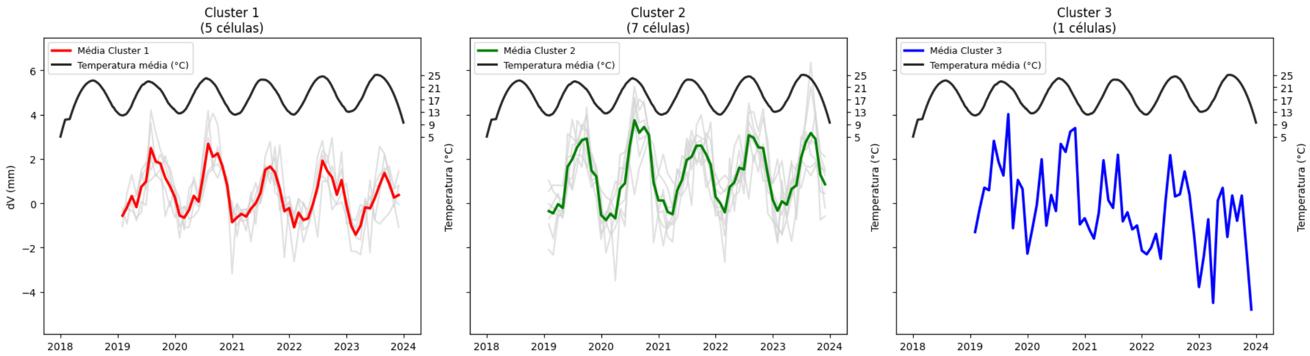


Figure 9: Mean vertical displacement (dV) per cluster with smoothed temperature (black line).

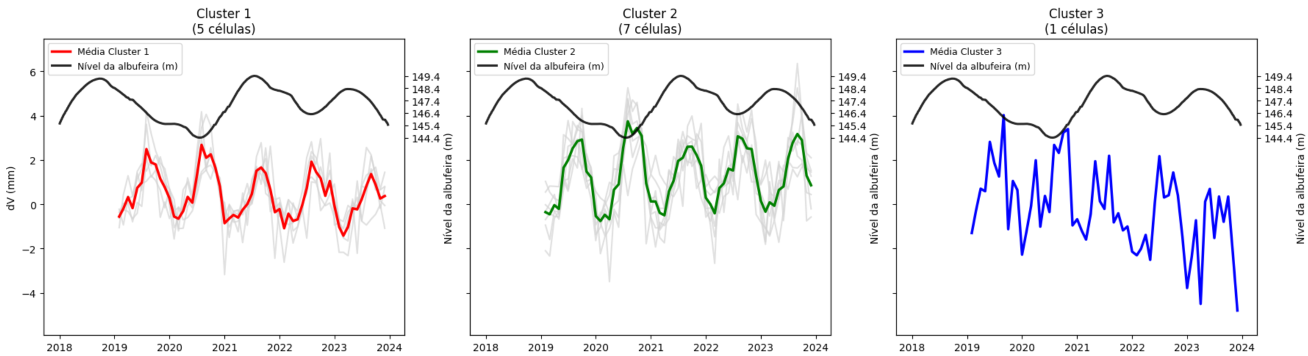


Figure 10: Mean vertical displacement (dV) per cluster with reservoir water level (black line).

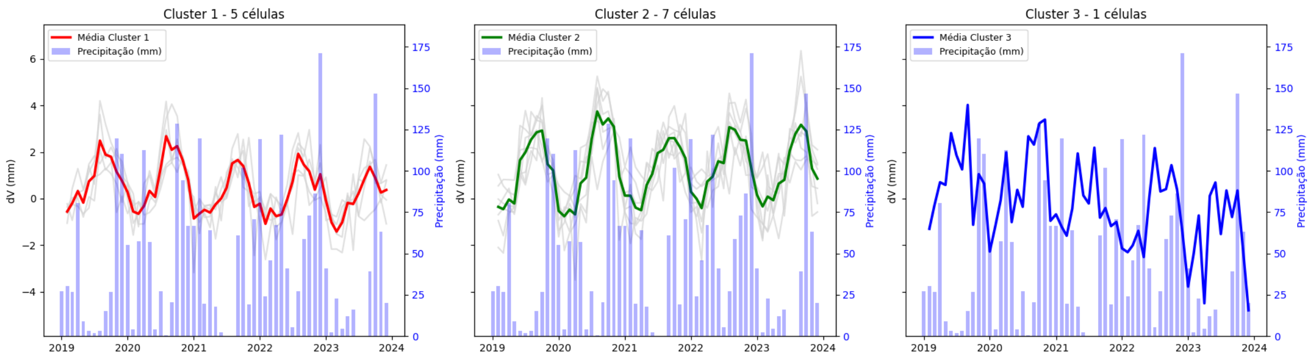


Figure 11: Mean vertical displacement (dV) per cluster with precipitation (blue bars)

processes.

DTW measures the similarity between two sequences $X = (x_1, \ldots, x_n)$ and $Y = (y_1, \ldots, y_m)$ by finding the optimal alignment path w that minimizes the cumulative distance:

$$\mathrm{DTW}(X, Y) = \min_{w} \sum_{(i,j)\in w} \|x_i - y_j\| \tag{8}$$

The hierarchical algorithm begins by treating each time-series as an individual cluster and successively merges them based on their DTW-based distance, according to a chosen linkage criterion (e.g., average or Ward's method). The resulting dendrogram provides a visual representation of cluster hierarchy, revealing groups of time-series that share similar deformation dynamics. The dendrogram was cut at the level that maximized the Silhouette Score, ensuring well-separated and physically meaningful groupings.

This combined DTW–Hierarchical approach offered:

- A detailed visualization of relationships among deformation patterns,
- The ability to detect gradual transitions between deformation regimes, and
- Independent validation of the K-Means results.

2.2.4 Temporal Decomposition of Clustered Time-Series

To further explore the temporal dynamics of the displacement signals, STL decomposition was applied to the cluster-mean series from both K-Means and DTW-Hierarchical clustering. Trend, seasonal, and residual components are presented for each cluster.

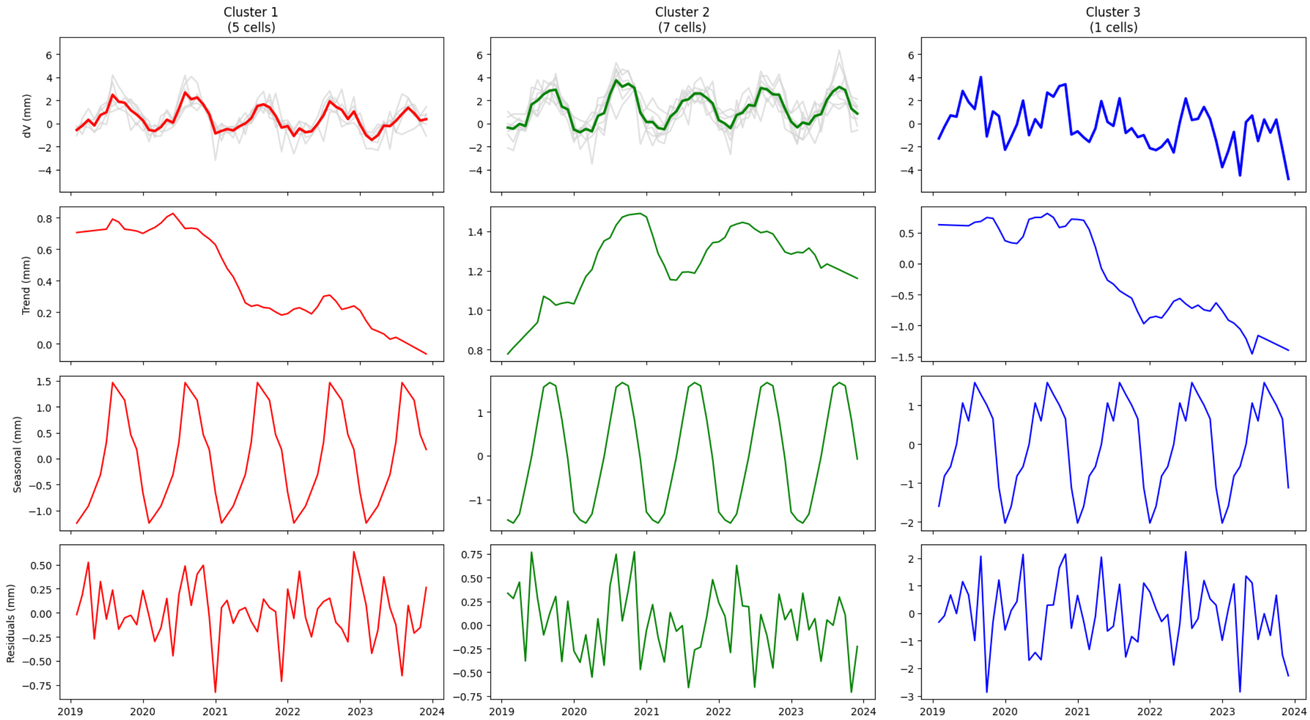


Figure 12: STL decomposition of mean vertical displacement (dV) per cluster: trend (top), seasonal component (middle), residuals (bottom). Cluster colours correspond to K-Means results.

Overall, the use of K-Means and DTW-based Hierarchical Clustering enabled a comprehensive understanding of deformation behaviour—from dominant regional trends to subtle local variations—supporting a more robust interpretation of the InSAR-derived displacement field.

3 Conclusion

Summarize key points and future work.

Data availability:

The datasets analysed during the current study are publicly available. Interferometric data are provided by the European 470 Ground Motion Service (EGMS) (https://egms.land.copernicus.eu/),

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